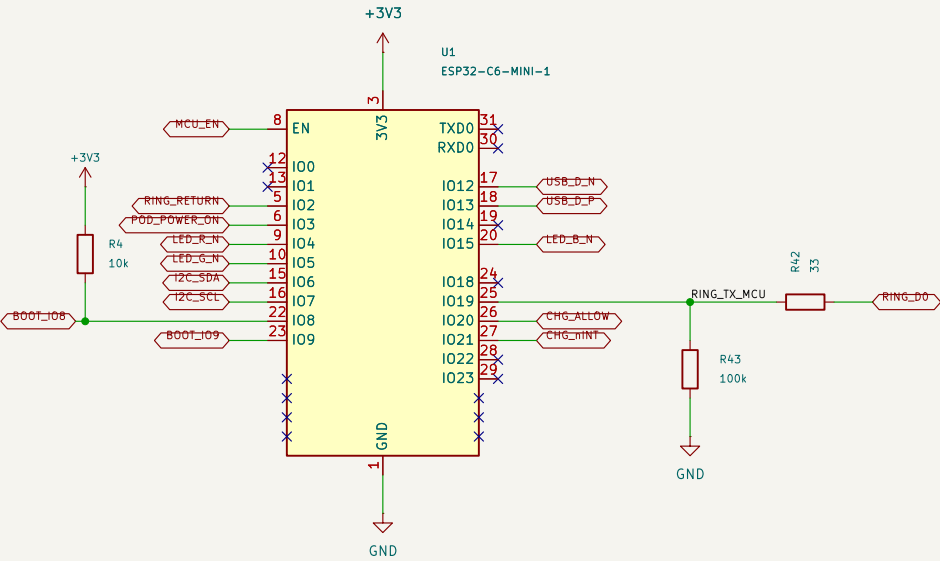


HAPTIC BRACELET / CONTROLLER DAUGHTERBOARD

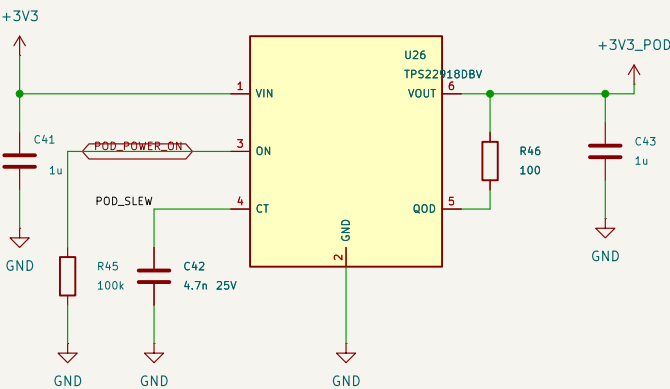
250 kbaud ring: ESP TX -> J101 -> universal pod 0 -> ... -> pod 7 -> RETURN -> ESP RX.
8 identical pod PCBAs + 1 controller daughterboard; protected batteries/NTCs on pods 1-7 only.



TX pulldown keeps the ring quiet while U1 resets.
Before pod power OFF: detach UART, TX low;
RX input with pull-up disabled. No signal isolators.

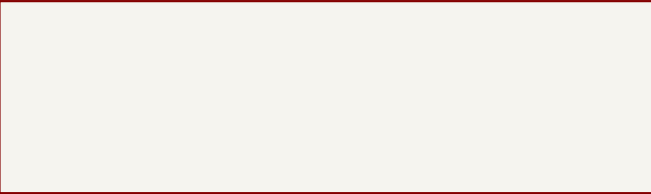
GPIO19 ring TX / GPIO2 ring RX
GPIO18 spare (no connect)
GPIO3 high powers pods
GPIO6/7: charger I2C only
GPIO0/1/14/22/23 spare (USB controls removed)
UART0 pads removed; native USB handles programming/debug

Pod recovery: cycle +3V3_POD through U26.
No shared reset wire; local ICE reset only.



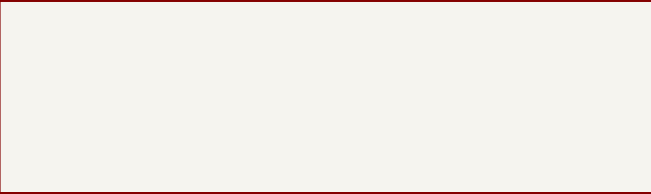
POD SUPPLY: default OFF; ESP, USB and charger stay powered.
CT controls ramp; Q0D discharges the ring after shutdown.
Do not enable UART idle-high until POD_3V3 has settled.
Disable charging before removing required pod temperature sensing.

Stacked universal pod interface



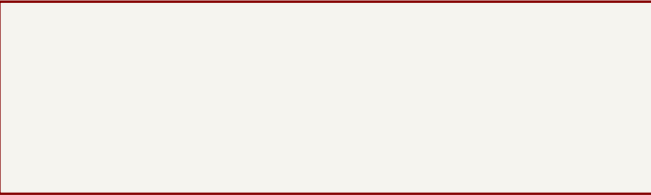
File: pod-interface.kicad_sch

Charger and regulated supply



File: power.kicad_sch

USB-C data and power



File: usb.kicad_sch

ESP reset, boot and decoupling



File: controller-support.kicad_sch

One wrist / ring pod schematic draft

Sheet: /
File: main.kicad_sch

Title: **Haptic bracelet – stacked controller**

Size: A3 Date: 2026-09-14

KiCad E.D.A. 10.0.1

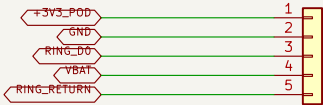
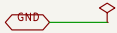
Rev: 0.5

Id: 1/5

CONTROLLER DAUGHTERBOARD / FIVE-CONTACT STACK INTERFACE

J101 connects pin-for-pin to J1 on universal pod 0.
The existing five-pad footprint is retained until the mechanical stack is set.

External supply entry
through J101: GND and VBAT



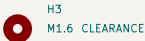
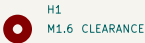
J101
STACK TO POD 0

- 1 Switched 3.3 V: daughterboard -> all eight pods
- 2 Common ground
- 3 ESP TX through R42 -> pod 0 RX
- 4 Protected VBAT bus: seven cells <-> charger
- 5 Last pod TX return -> ESP RX

SYSTEM ASSEMBLY
One controller daughterboard; eight copies of satellite/satellite.kicad_pro.
Pod 0 sits beneath this board; no battery and no external NTC fitted.
Pods 1-7 each carry one protected pouch cell and its external NTC.
The pod PCBA and component population are identical in all positions.

CHAIN AND RETURN
Daughterboard J101 -> pod 0 J1. Each pod J2 -> next pod J1.
Pods 0-6: JP1 NORMAL (1-2). Pod 7: cut 1-2, bridge 2-3 END.
Pod 7 J2 is uncabled. The clasp gap remains unwired.

POWER AND TEMPERATURE
All eight M2003/DRV2625 nodes share switched +3V3_POD.
Only pod 0 battery/NTC sockets are unused; keep its sensing/fuse circuitry.
Firmware expects seven required cell sensors on pods 1-7.
Disable charging if a required sensor is missing, stale or out of range.
Seven-cell charge/load limits still require firmware configuration.



H1-H4: four corner M1.6 mounts.
20 x 46 mm PCB faces / TPU spacer.
Standoff height and screw length TBD.

One wrist / ring pod schematic draft

Sheet: /Stacked universal pod interface/
File: pod-interface.kicad_sch

Title: Controller to universal pod stack

Size: A4	Date: 2026-09-14	Rev: 1.0
KiCad E.D.A. 10.0.1		Id: 2/5

